Vending Machine – Design Document

Design Document

Vending Machine automates retail transactions without human intervention by managing product selection, multi-mode payments, mechanical dispensing, and real-time inventory tracking.

Project Layers

- **Entities**: Core database models and state enums.
- **Data**: API request and response contracts, Thread-safe in-memory store with pre-seeded inventory.
- **Services**: Third-party payment client, hardware dispenser mock, and core vending orchestrator.
- **Controllers**: RESTful API endpoints.

Domain Entities

Entity	Key Properties	Purpose
VendingMachine	machine_id, location, status ENUM, last_OperatedTime:	Stores physical machine location, status and last OperatedTime.
Product	product_id, name, price, category	Defines the purchasable products, including display name, unit selling price, and product category.
Inventory	inventory_id,machine_id, product_id, slot_number, quantity	Maps products to physical machine slots (1–50) and tracks real-time available unit counts (0–20).
Transaction	transaction_id, machine_id, product_id, total_amount, status, created_at	Tracks the end-to-end order lifecycle (IDLE, SELECTED, PAID, DISPENSED, FAILED, CANCELLED) and enforces idempotency.
Payment	payment_id, transaction_id: , payment_method, amount_paid, change_returned, payment_status	Records payment method (CASH, CARD, UPI), amount tendered, change returned, and settlement state (PENDING, SUCCESS, FAILED, REFUNDED).
LocalStoreOrder	Id, OrderId, OrderStaus, Time	It store data in local Store and delete after transaction sucessfull. This will use for any issue while processing transaction
VendingMachineUser	UserId,Name,UserType, IsActive,UserTimeOutInSeconds	

Services Or Operations

Service	Method Name	Return Type	Purpose
Product & Inventory	getProducts	List<Product>	Retrieves the full catalog of products configured for a specific machine.
Product & Inventory	getInventory	List<Inventory>	Fetches real-time stock quantities and slot mappings for UI availability.
Product & Inventory	updateStock	BOOLEAN	Decrements stock post-dispense or increments stock during restocking.
Transaction	createTransaction	Transaction	Initializes an order, locks the slot, and sets state to SELECTED.
Transaction	cancelTransaction	BOOLEAN	Aborts an active order, releases the slot lock, and triggers a refund if paid.
Payment	processPayment	Payment	Validates inserted cash or authorizes digital funds (Card/UPI) via gateway.
Payment	refundPayment	BOOLEAN	Reverses a completed payment upon user cancellation or dispense failure.
Dispense & Hardware	dispenseProduct	BOOLEAN	Triggers hardware motor via GPIO/Serial and verifies release via drop sensor.
Dispense & Hardware	returnChange	BOOLEAN	Commands the cash dispenser to eject exact change when cash exceeds price.

Functional Requirements:

- Users should be able to view available products and their prices.
- Users should be able to select a product for purchase.
- The machine should accept supported payment methods such as cash, card, or digital payment.
- The system should verify that enough balance has been provided before dispensing the product.
- The machine should dispense the selected product and return change when required.
- The inventory should be updated after every successful purchase.

- The system should handle out-of-stock products, payment failures, cancellations, and refunds.

Non-Functional Requirements:

- **Reliability:** The machine should dispense the correct product after successful payment.
- **Consistency:** Inventory and payment status should always remain accurate.
- **Low Latency:** Product selection and payment processing should complete quickly.
- **Availability:** The machine should remain operational whenever products and required hardware are available.
- **Security:** Payment information and machine operations should be protected from unauthorized access.
- **Maintainability:** Products, prices, inventory, and payment components should be easy to update or replace.